

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
III B. TECH II SEMESTER

BATCH 2023-27	MINI PROJECT 2025-26	ACADEMIC YEAR 2025-26
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Batch Id : 23MPCS-B11

Title of the Project : A Multi-Agent Bee Colony Framework for Reliable User Behavior Prediction on Edge Healthcare Devices.

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Batch Id : 23MPCS-B11

Title of the Project : A Multi-Agent Bee Colony Framework for Reliable User Behavior Prediction on Edge Healthcare Devices.

Objectives

1. To design a Multi-Agent Bee Colony (MABC)–based framework for accurate user behavior prediction directly on edge healthcare devices, enabling real-time analysis without relying on cloud processing.
2. To improve reliability and robustness of behavior prediction by integrating swarm intelligence algorithms with sensor data collected on edge-enabled smart healthcare devices.
3. To reduce latency and enhance decision-making in healthcare IoT systems by performing behavior prediction locally at the edge.
4. To optimize computational efficiency, memory usage, and energy consumption of edge healthcare devices using an adaptive MABC optimization approach.
5. To validate the proposed edge-based MABC framework through experiments on real-time edge healthcare environments and benchmark datasets, measuring accuracy, speed, and resource efficiency.

Outcomes

1. A reliable trust-prediction model that identifies trustworthy and untrustworthy users in edge-based healthcare device environments.
2. Improved accuracy and stability of health-monitoring systems using MABC-optimized classification.
3. Enhanced security and data integrity for wearable and IoT-health devices by preventing misuse or false data entry.
4. Reduced false alarms and improved patient monitoring, leading to better clinical decision-making.
5. A robust edge-AI framework capable of processing and validating health data efficiently with minimal delays.

ABSTRACT

With the rapid adoption of smart healthcare devices, accurately predicting user behavior is crucial for personalized care and timely interventions. This project proposes a Multi-Agent Bee Colony (MABC) based framework to model and predict user behavior in smart healthcare systems. The framework leverages the collaborative intelligence of multiple agents, inspired by the foraging behavior of bee colonies, to optimize prediction accuracy while maintaining system reliability. By analyzing user interaction patterns, physiological data, and device usage, the MABC framework dynamically adapts to changing user habits, ensuring proactive healthcare monitoring. Experimental results demonstrate that the proposed approach improves prediction accuracy, enhances user trust, and supports real-time decision-making in healthcare applications.

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Date of Submission: 01-12-2025

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